TCC 2026

A Low-Communication Garbled RAM from Homomorphic Secret Sharing



Chase Fickes



Jinye He



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Photo taken during TCC





Photo taken during TCC



My colleague: Dr. Bear





Photo taken during TCC





My colleague: Dr. Bear

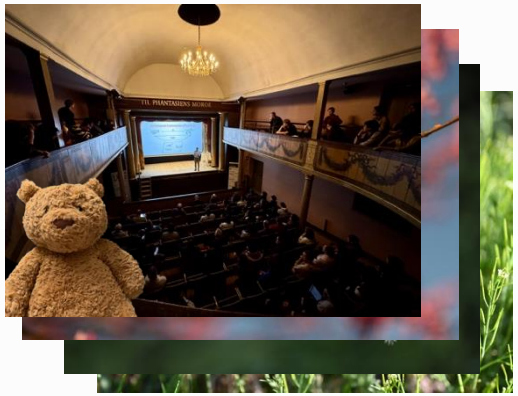




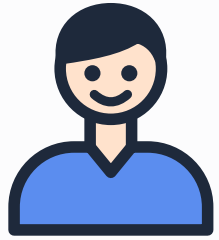
Ten years of my photos



My colleague: Dr. Bear



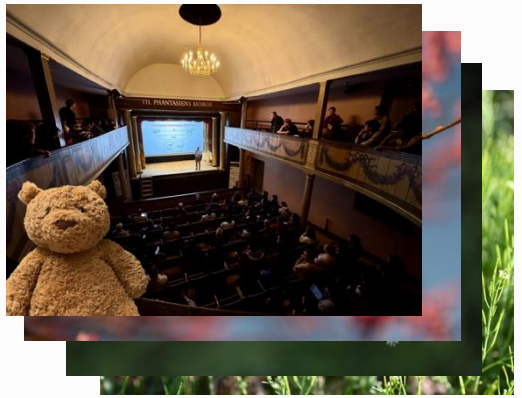




Cloud server and storage











Yao's Garbled Circuit [Yao'86]

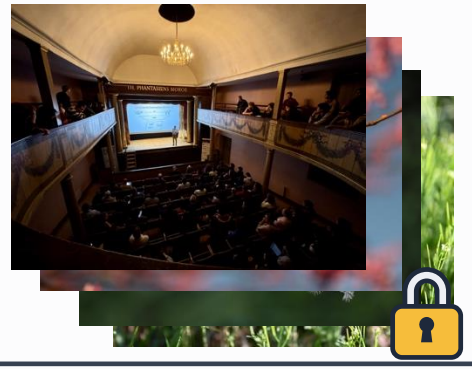
Upload all my photos



Client

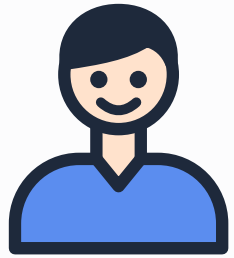
Garbler





Upload the encoded photos





Cloud server and storage

Cloud server edits the photos and sends back



Client

Garbler



1. Find the TCC photos
2. Add Dr. Bear to them

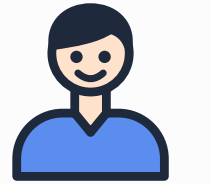
Garbled edit program $\tilde{P}$



Encrypted edited photos



Cloud storage



Evaluator

Decode and get the edited photos



Client

Garbler





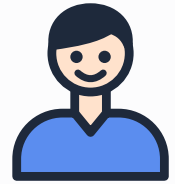
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Encrypted edited photos

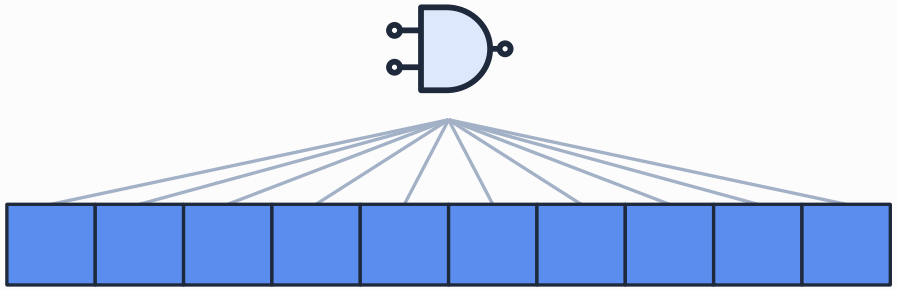


Cloud storage



Evaluator

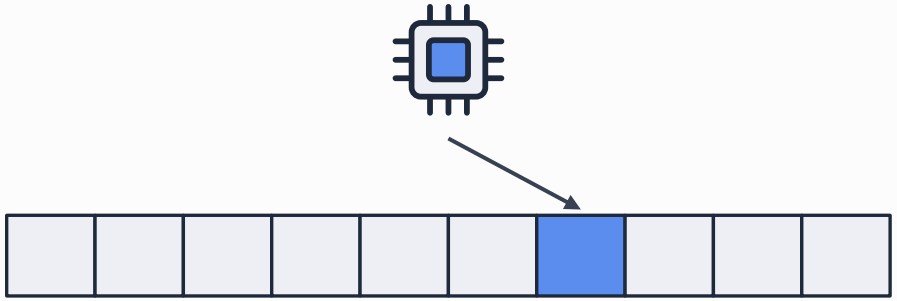
Circuits vs. RAM Programs



Circuit

Fixed wiring: every access touches all N words

size $\approx T \cdot N$



RAM program

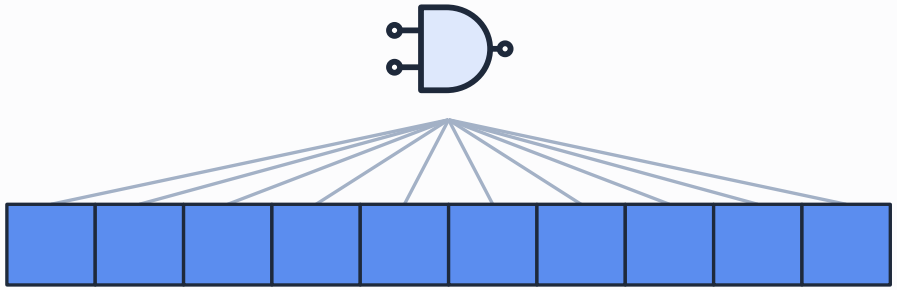
Random access: each step touches just one word

T steps

Naive baseline: compile a RAM program P into a circuit

- T: program running steps
- N: memory size
- W: word size
- Communication: $\boldsymbol{O(TNW \cdot \lambda)}$

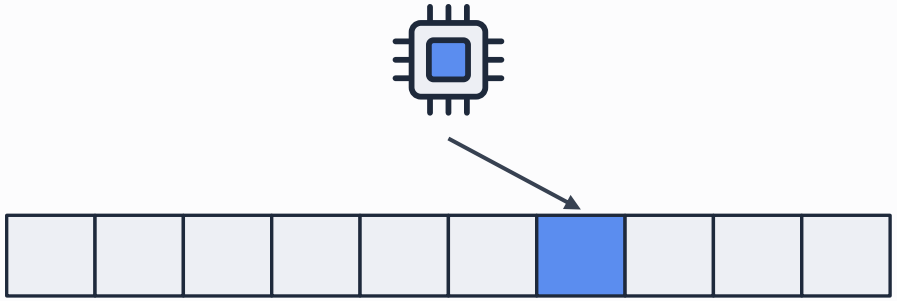
Circuits vs. RAM Programs



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RAM program

Random access: each step touches just one word

T steps

Adding Dr. Bear to my 50 TCC photos, among 10,000 photos in the cloud
GC touches every pixel of all 10,000 photos: ⇒ **more than 8 days** of communication (1 Gb/s)

Cannot complete within this TCC week!

GRAM can save TCC photo editing!

Construction	Communication (bits)	Assumption
Tri-state Circuits [Heath-Kolesnikov-Ostrovsky'23]	$O(\lambda \cdot TW \log N \cdot \text{loglog } N)^*$	OWF
Garbled Merge [Liu-Liu-Luo-Heath'26]	$O(\lambda \cdot TW \log N \cdot \omega(1))^{**}$	RO
Zebra [Gu-Ghoshal-Shi'26]	$O(TW \cdot (\log N + \lambda))^{***}$	DCR+RO

Reducing N to $\log N$: **8 days** → **3 hours** editing work!

*, ** Adapted from [HKO'23] and [LLLH'26]. The communication bounds assume sufficiently large word size $W \geq \Omega(\log^2 N)$.

*** [GGS'26] achieves $O(TW \log N)$ for arithmetic RAM. We focus on boolean RAM here.

Existing works pay a multiplicative λ

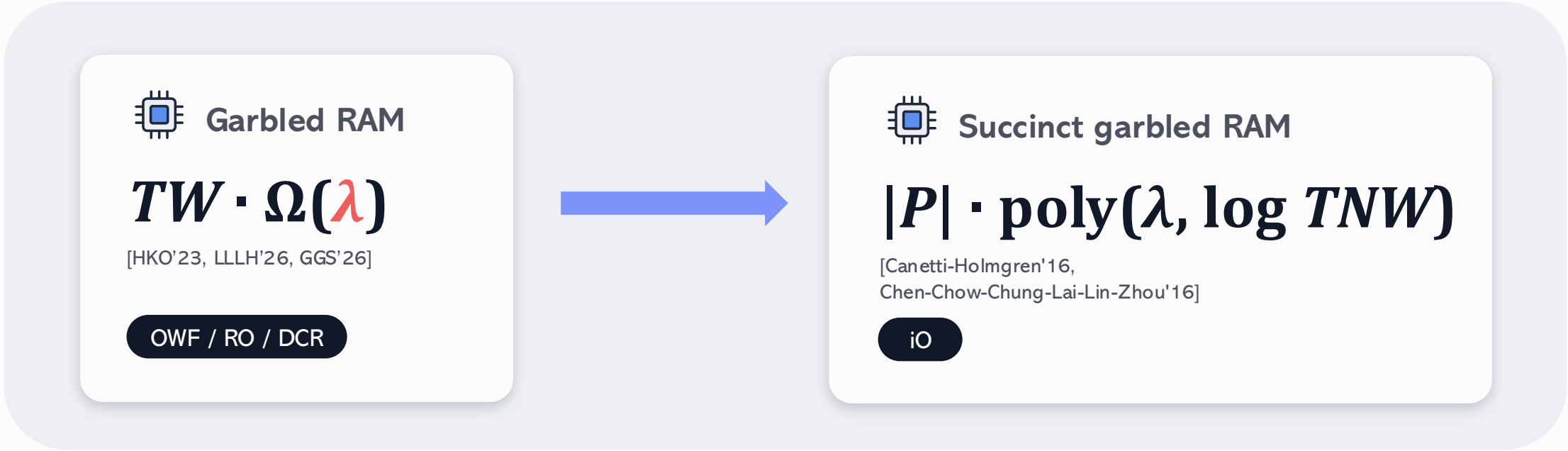
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? Can we get rid of this λ ? **3 hours** → 1.5 mins (~100× faster)

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Assuming IO can partially answer



Can we get better GRAM without IO?



Can we borrow technique from succinct garbled circuit?

Yao's garbled circuits [Yao'86]

$|C| \cdot \Omega(\lambda)$

≈ λ bits per gate

Succinct garbled circuits

$|C| + \mathbf{poly}(\lambda)$

≈ 1 bit per gate

from DDH-, DCR- or RLWE-type assumptions

[Liu-Wang-Yang-Yu'25, Meyer-Orlandi-Roy-Scholl'25, Ishai-Li-Lin'25, Li-Lin-Lu'26]

Garbled RAM

$TW \cdot \Omega(\lambda)$

[HKO'23, LLLH'26, GGS'26]

OWF / RO / DCR

?

Succinct garbled RAM

$|P| \cdot \mathbf{poly}(\lambda, \log TNW)$

[Canetti-Holmgren'16, Chen-Chow-Chung-Lai-Lin-Zhou'16]

iO

Our Result

Achieve $O(T \cdot W \cdot \log N) + W^2 \cdot \mathrm{poly}(\lambda)$ for large words

Theorem (informal). Assuming circular-power DDH or circular-power RLWE, there is a garbled RAM for boolean RAM programs with communication

General

$$O(T \cdot (W + \lambda) \cdot \log N) + W^2 \cdot \mathrm{poly}(\lambda)$$

λ added to W, not multiplied

one-time, independent of T

Large words $W \geq \lambda$

$$O(T \cdot W \cdot \log N) + W^2 \cdot \mathrm{poly}(\lambda)$$

$\approx \log N$ bits per word-bit

λ: bit-length of a group or ring element

The only overhead left is ORAM's $\log N$

	Communication (bits)	Round Trips
Oblivious RAM [Goldreich-Ostrovsky'96, Wang-Chan-Shi'15 Asharov-Komargodski-Lin-Nayak-Peserico-Shi'20]	$O(TW \cdot \log N)$	T
Prior garbled RAM [HKO'23, LLLH'26]	$O(\lambda \cdot TW \cdot \log N)$	1
Our garbled RAM Word size $W \geq \lambda$	$O(TW \cdot \log N)$ $+ W^2 \cdot \mathrm{poly}(\lambda)$	1

The best of both worlds!

Most important building block——Route

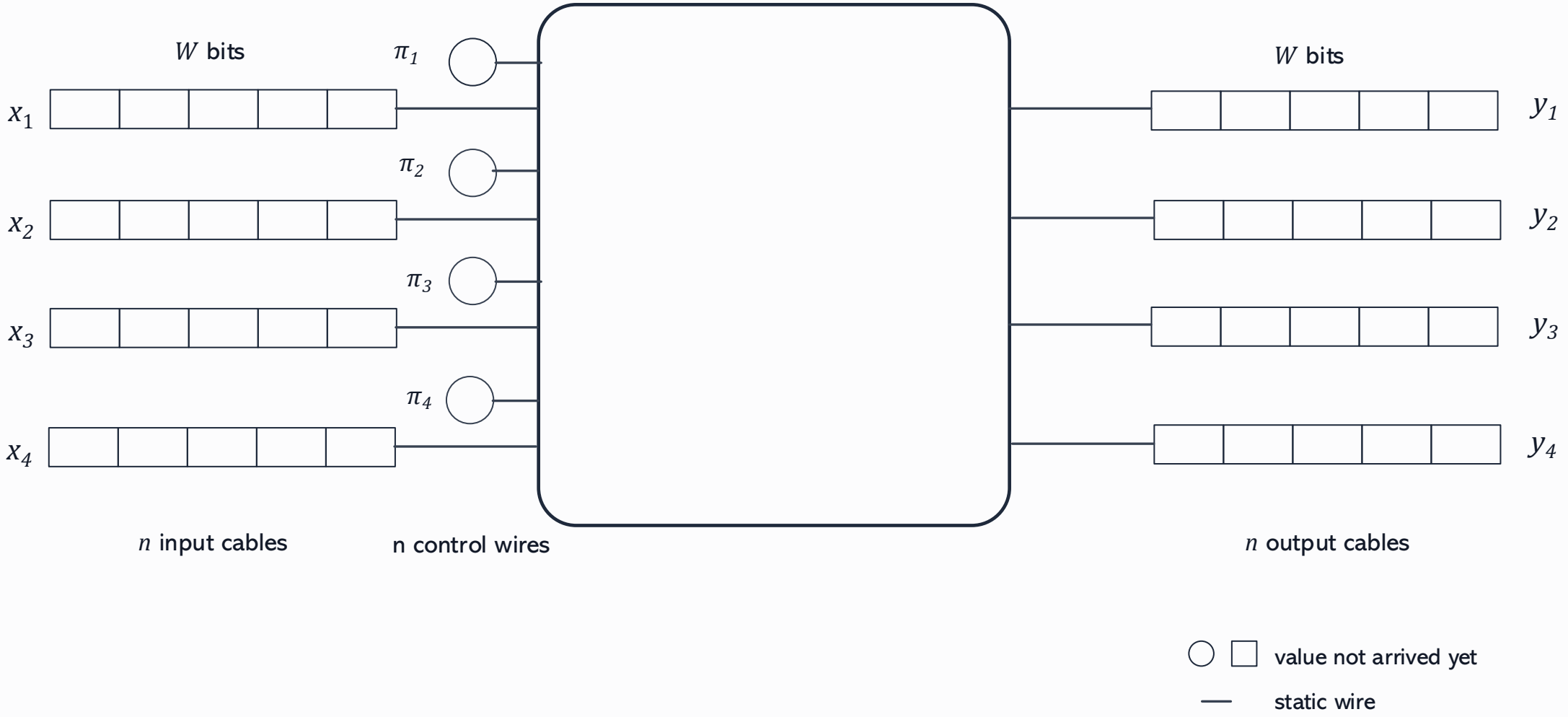
Route: the key circuit gadget in [HKO'23, GGS'26, LLLH'26]

Most important building block——Route

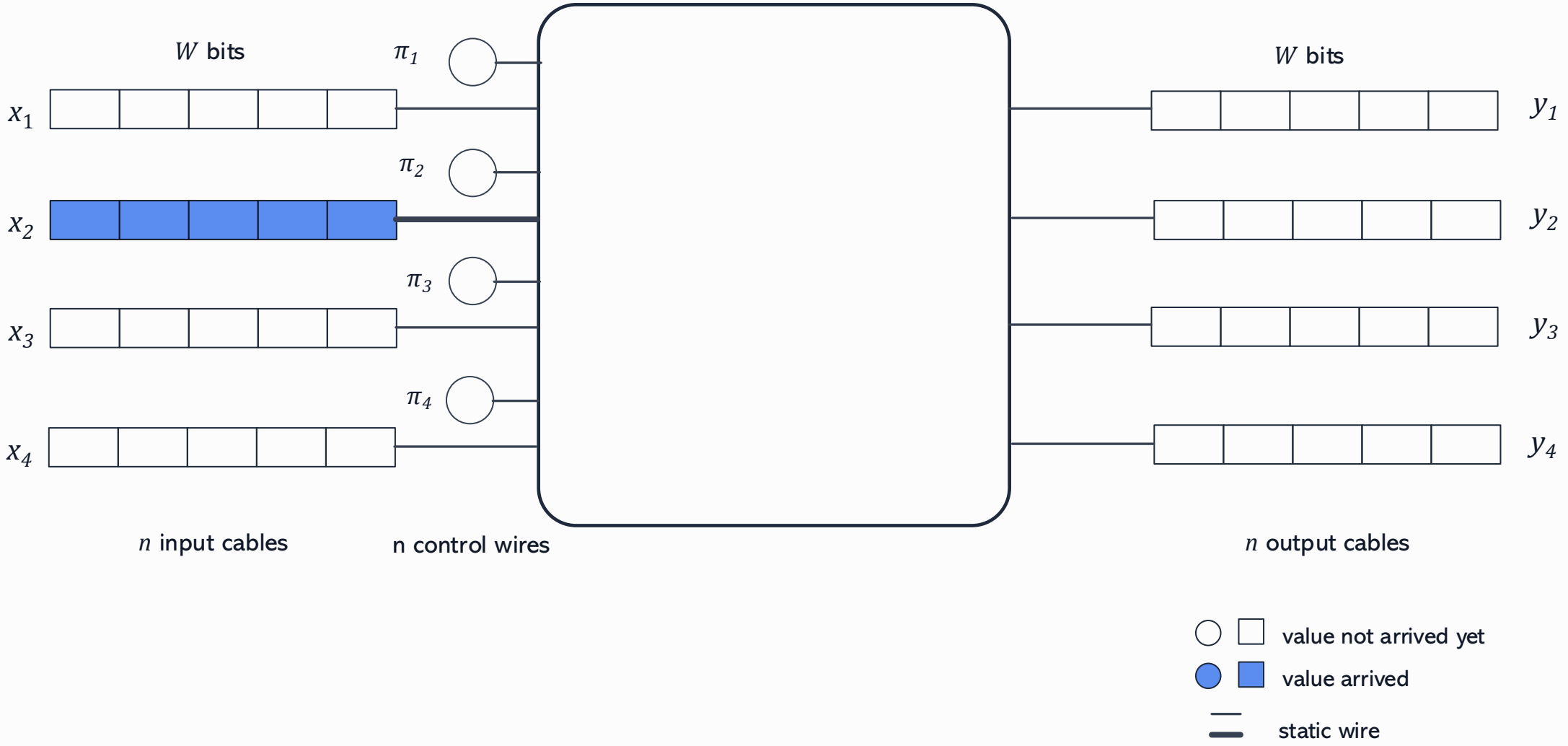
Route: the key circuit gadget in [HKO'23, GGS'26, LLLH'26]

Better garbled Route $\Rightarrow$ Better garbled RAM

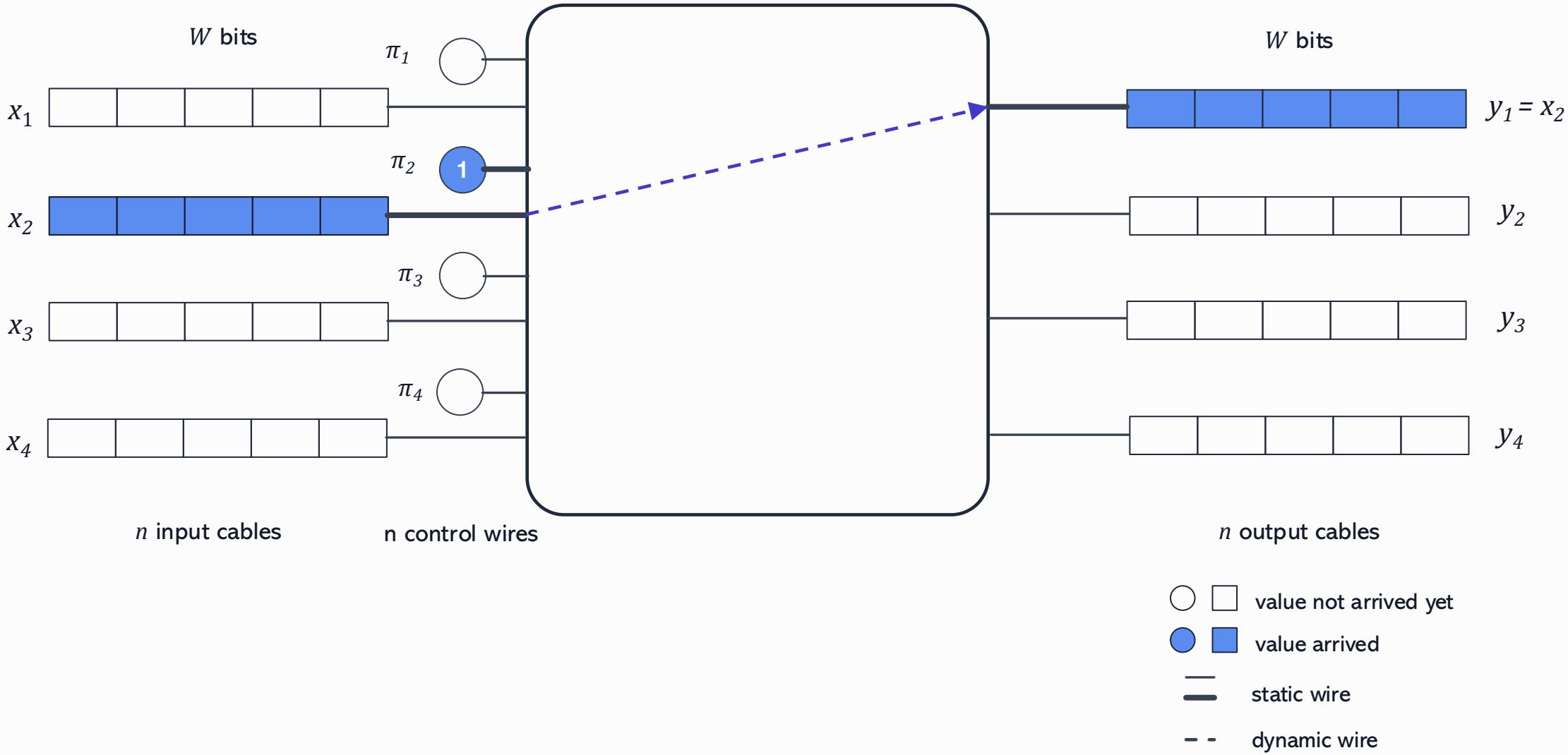
Functionality of Route



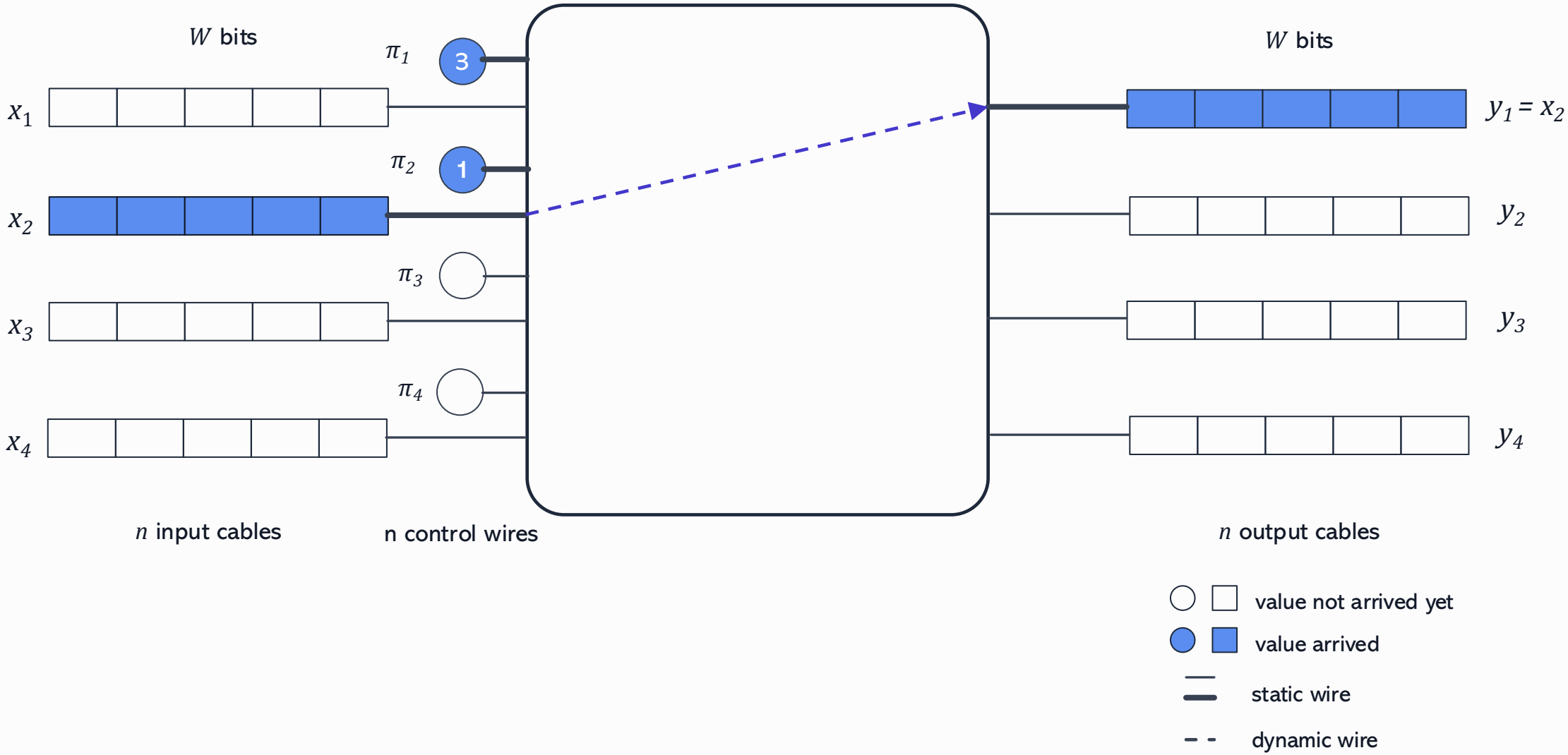
The value carried by cable *2* arrives



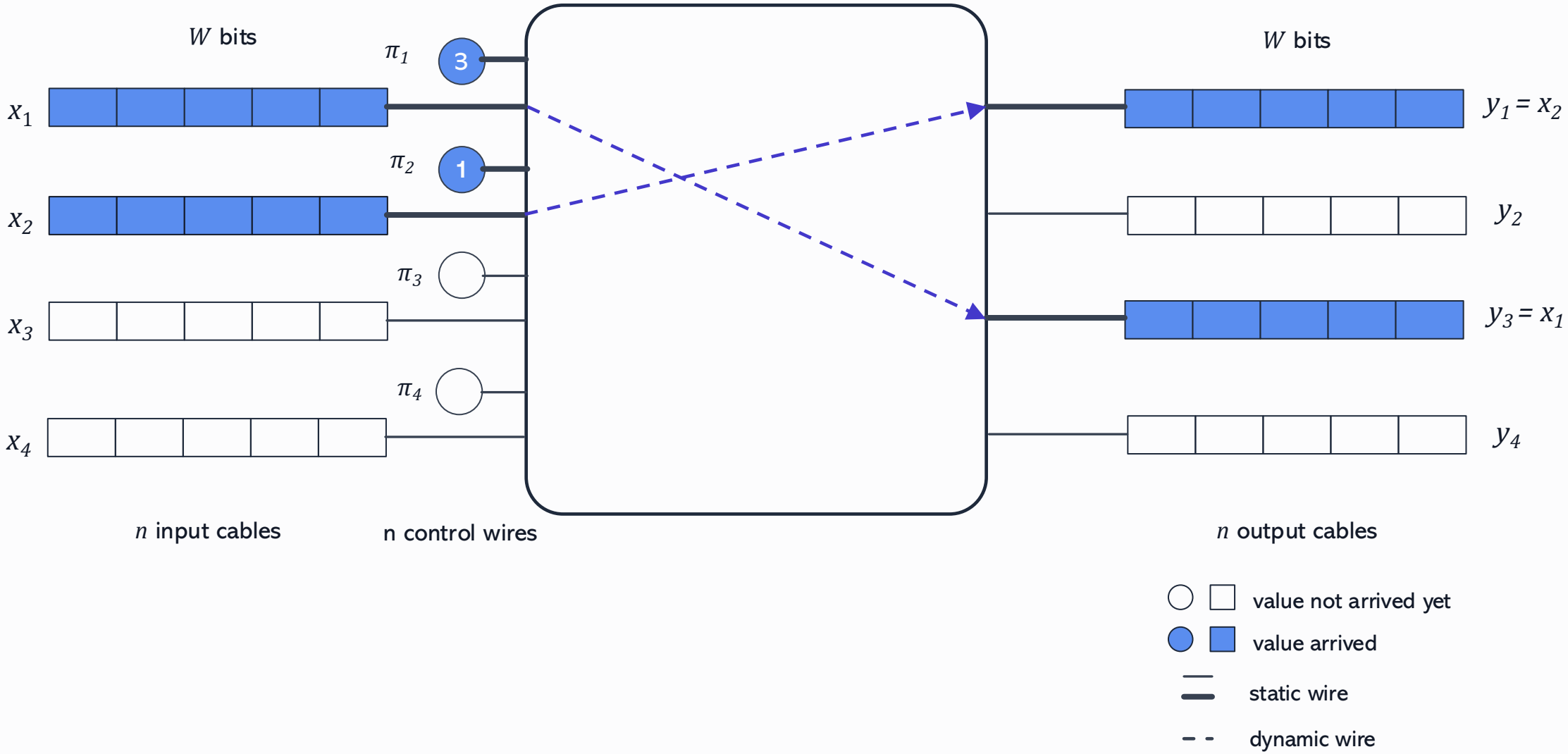
Cable *2* fires once x_2 and π_2 arrive, without waiting for the others



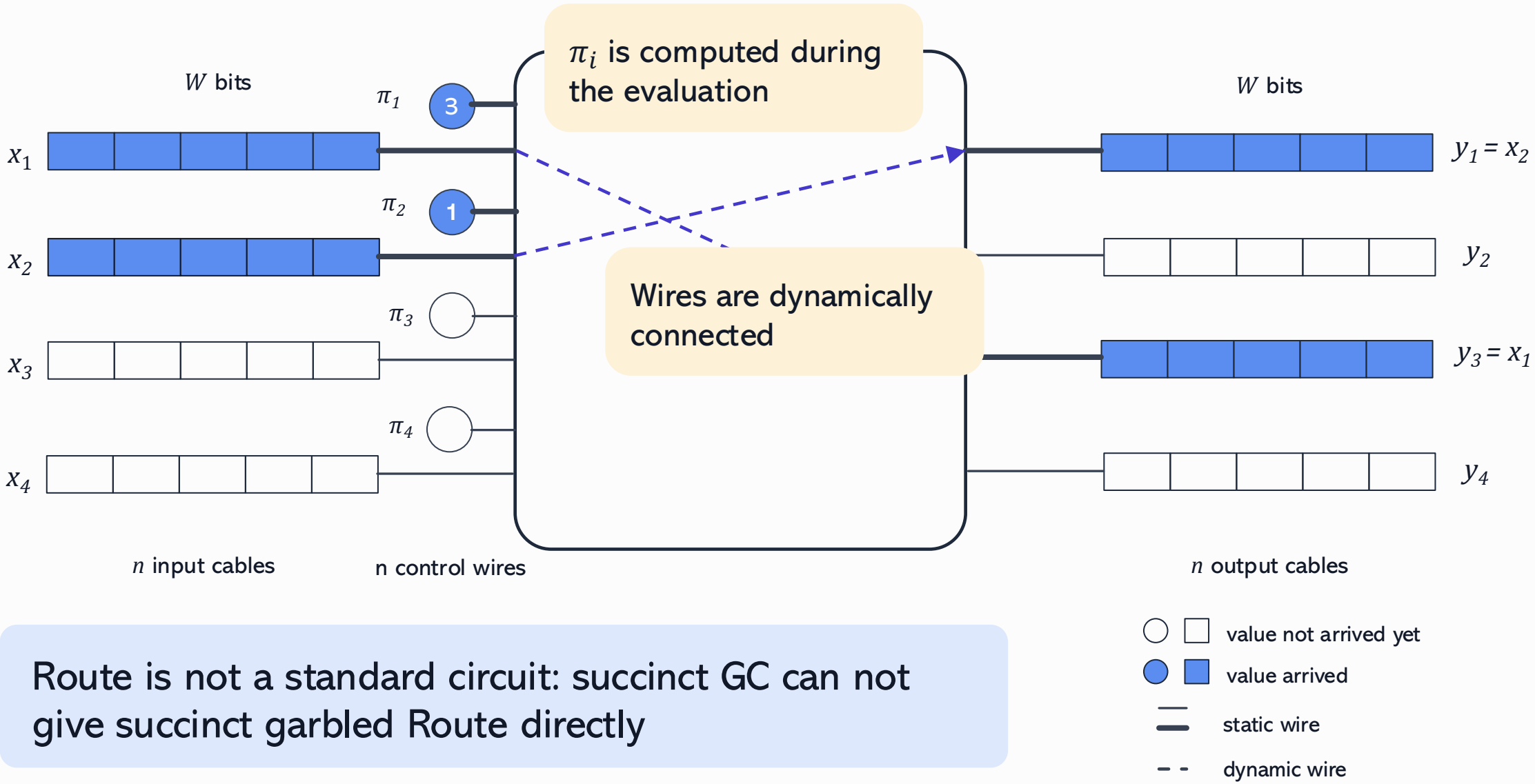
The control wire π_1 arrives



Cable *1* fires once x_1 and π_1 arrive



Cable *1* fires once x_1 and π_1 arrive



Route is not a standard circuit: succinct GC can not give succinct garbled Route directly

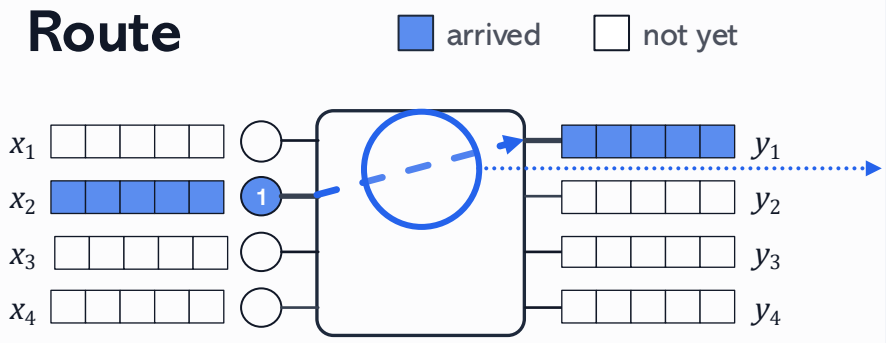
Communication of garbled Route

	Bits per cable	Assumption
Garbled Route [GGS'26]	$O(W + n \cdot \lambda)$	DCR+RO
Garbled Merge [LLLH'26] · a special case of Route	$O(W \cdot \lambda)$	RO
Our garbled Route	$O(W + \lambda)$ $+ W^2 \cdot \mathrm{poly}(\lambda)$ one-time	CP-DDH / CP-RLWE

λ is added, not multiplied — and independent of n

Idea 1: garble a GRF

Generate labels through PRF





Garbler

labels of x_2



W input labels
λ bits each

$\mathbf{PRF_{sk}(in, 2, t)}\ \forall\ t \in [W]$

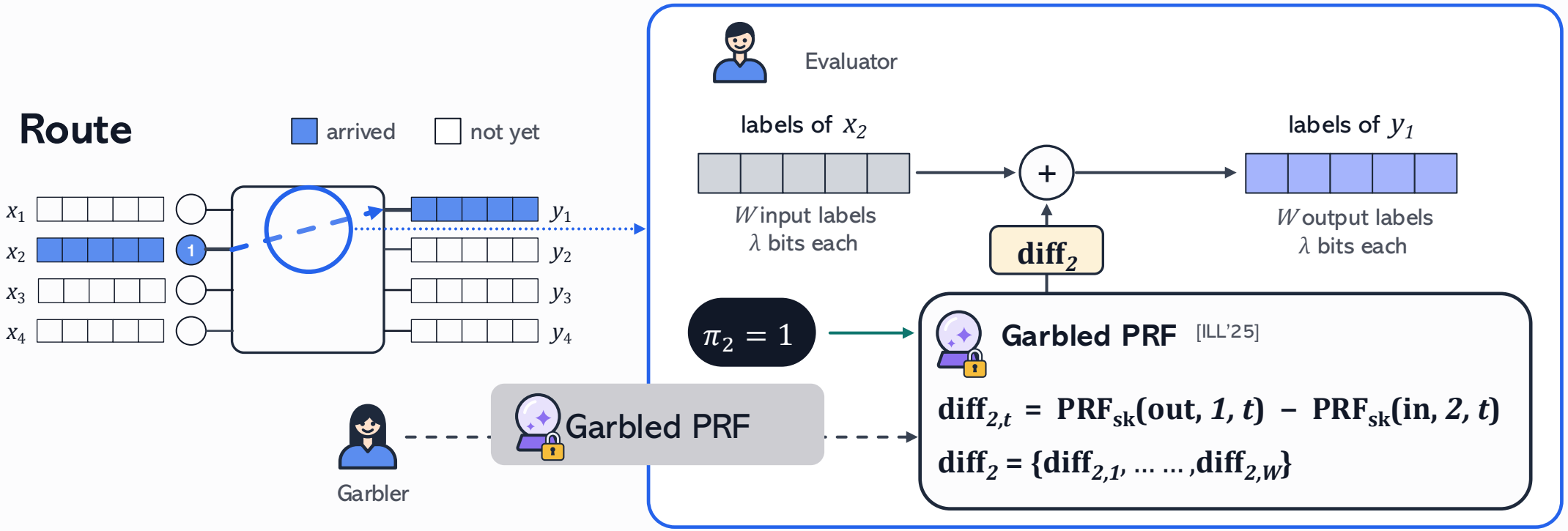
labels of y_1



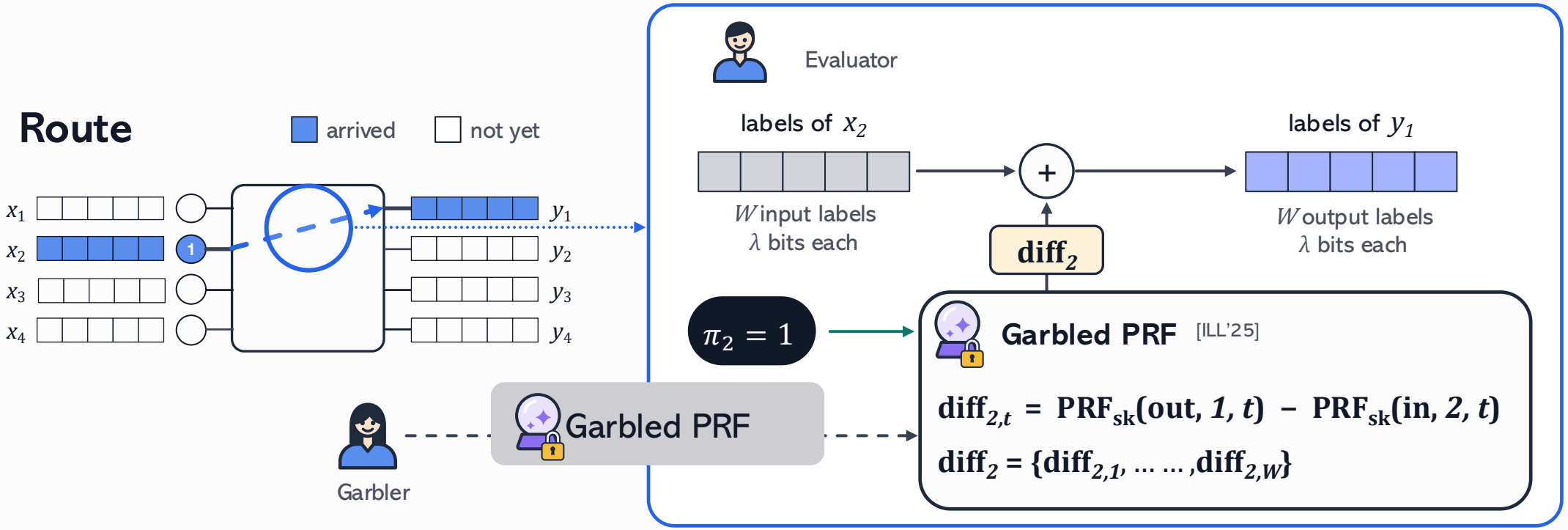
W output labels
λ bits each

$\mathbf{PRF_{sk}(out, 1, t)}\ \forall\ t \in [W]$

Compute the labels through garbled PRF



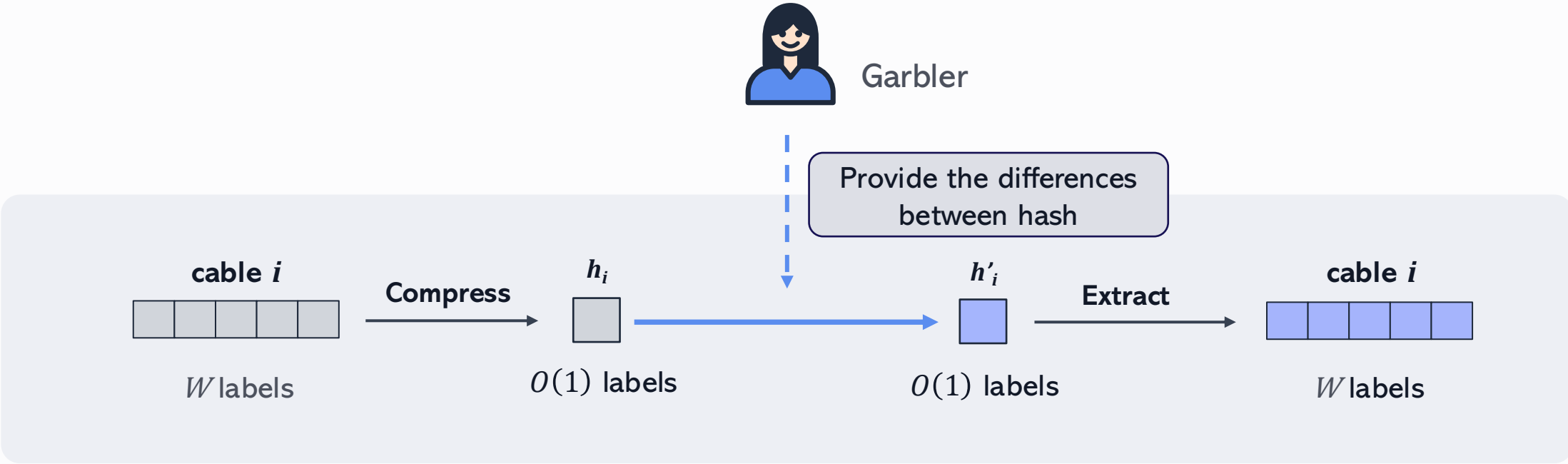
Correct, secure, but not efficient



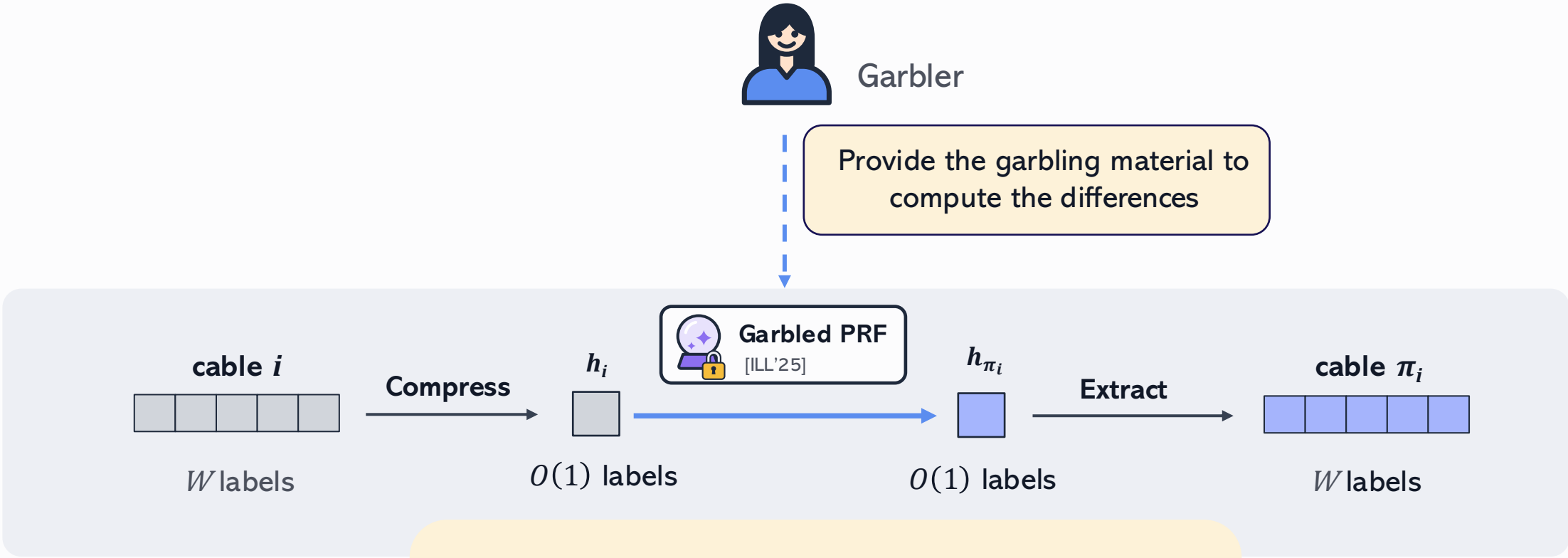
Per cable: delivering $\mathbf{diff}_i$ (W group/ring elements) costs $O(W \cdot \lambda)$ bits

Idea 2: Compress the labels

Fixed Factor Hashing [LLL'26]



Using Fixed Factor Hashing in our work

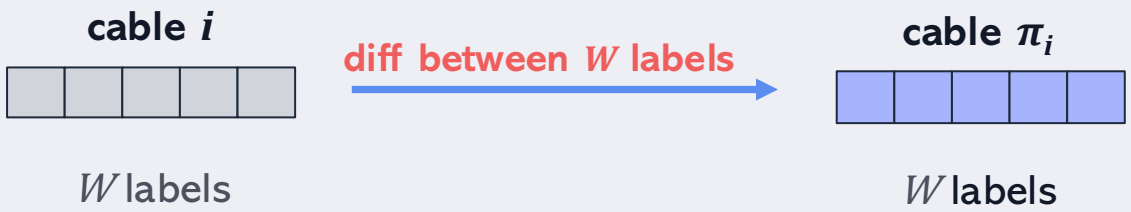


$$h_i = \mathrm{PRF}_{\mathrm{sk}}(\mathrm{in}, i) \qquad h_{\pi_i} = \mathrm{PRF}_{\mathrm{sk}}(\mathrm{out}, \pi_i)$$

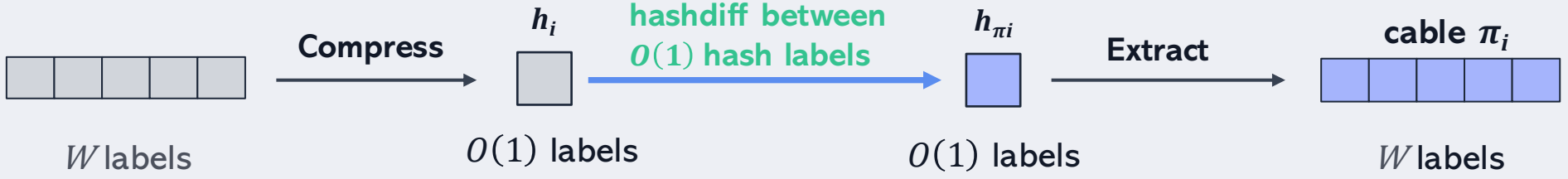
$$\mathrm{hashdiff}_i = \mathrm{PRF}_{\mathrm{sk}}(\mathrm{out}, \pi_i) - \mathrm{PRF}_{\mathrm{sk}}(\mathrm{in}, i)$$

Save W factor in Route

Strawman construction



Final construction



Per cable communication cost: $O(W + W \cdot \lambda)$ bits $\rightarrow$ $O(W + \lambda)$ bits





Questions?

Thank you!

eprint.iacr.org/2026/1998